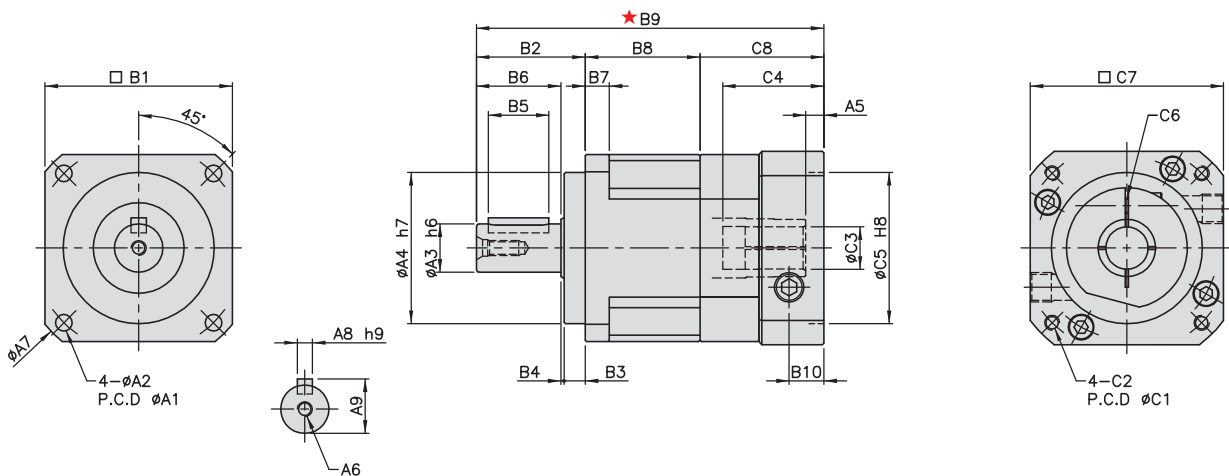
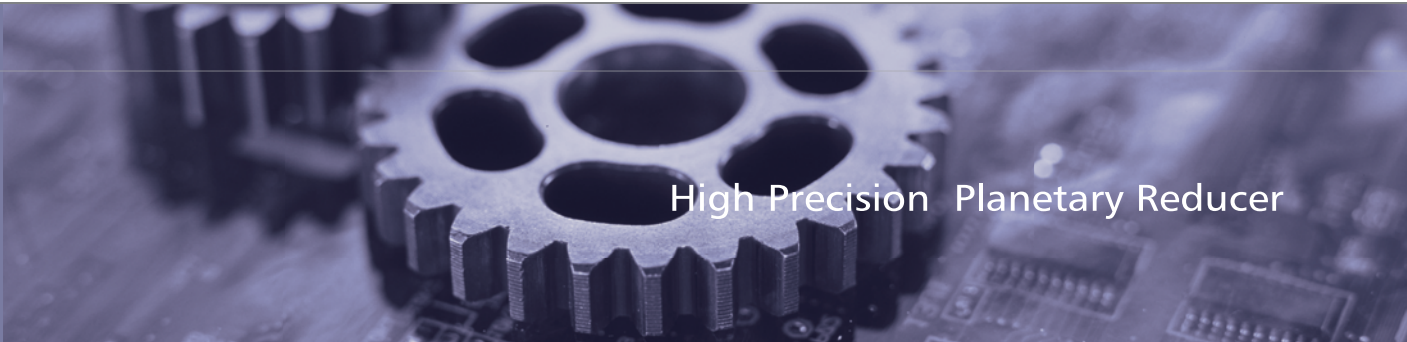
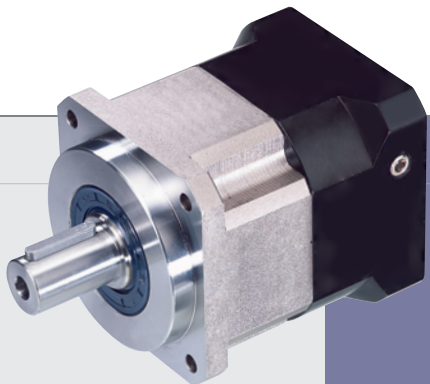


MODEL : KSB

Single Reduction  
RATIO : 3.4.5.6.7.8.9.10



■ Mass Moments of Inertia (kg · cm²)

| Ratio | 44   | 62   | 90   | 120  | 142  | 180   | 220   |
|-------|------|------|------|------|------|-------|-------|
| 3     | 0.03 | 0.16 | 0.61 | 3.25 | 9.21 | 28.98 | 59.61 |
| 4     | 0.03 | 0.14 | 0.48 | 2.74 | 7.54 | 23.67 | 54.37 |
| 5     | 0.03 | 0.13 | 0.47 | 2.71 | 7.42 | 23.29 | 53.27 |
| 6     | 0.03 | 0.13 | 0.45 | 2.65 | 7.25 | 22.75 | 51.72 |
| 7     | 0.03 | 0.13 | 0.45 | 2.62 | 7.14 | 22.48 | 50.97 |
| 8     | 0.03 | 0.13 | 0.44 | 2.58 | 7.07 | 22.59 | 50.84 |
| 9     | 0.03 | 0.13 | 0.44 | 2.57 | 7.04 | 22.53 | 50.63 |
| 10    | 0.03 | 0.13 | 0.44 | 2.57 | 7.03 | 22.51 | 50.56 |

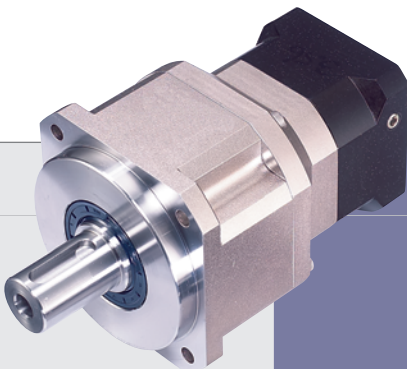
| Model No.                    | Unit       | Ratio | 44                                    | 62     | 90    | 120   | 142   | 180    | 220    |
|------------------------------|------------|-------|---------------------------------------|--------|-------|-------|-------|--------|--------|
| Rated Output Torque          | Nm         | 3     | 19                                    | 59     | 165   | 335   | 625   | 1206   | 2030   |
|                              |            | 4     | 16                                    | 51     | 146   | 300   | 555   | 1069   | 1804   |
|                              |            | 5     | 16                                    | 48     | 160   | 333   | 618   | 1189   | 2010   |
|                              |            | 6     | 15                                    | 45     | 151   | 311   | 583   | 1118   | 1911   |
|                              |            | 7     | 15                                    | 45     | 149   | 309   | 573   | 1108   | 1870   |
|                              |            | 8     | 14                                    | 43     | 143   | 298   | 553   | 1070   | 1824   |
|                              |            | 9     | 13                                    | 44     | 145   | 278   | 516   | 993    | 1694   |
|                              |            | 10    | 14                                    | 43     | 141   | 294   | 549   | 1059   | 1779   |
| Max. Output Torque           | Nm         | 3~10  | 3 Times of Rated Output Torque        |        |       |       |       |        |        |
| Rated Input Speed            | rpm        | 3~10  | 5,000                                 | 5,000  | 4,000 | 4,000 | 3,000 | 3,000  | 2,000  |
| Max. Input Speed             | rpm        | 3~10  | 10,000                                | 10,000 | 8,000 | 8,000 | 6,000 | 6,000  | 4,000  |
| Backlash PS                  | arc min    | 3~10  |                                       |        | ≤1    | ≤1    | ≤1    | ≤1     | ≤1     |
| Backlash P0                  | arc min    | 3~10  | ≤3                                    | ≤3     | ≤3    | ≤3    | ≤3    | ≤3     | ≤3     |
| Backlash P1                  | arc min    | 3~10  | ≤5                                    | ≤5     | ≤5    | ≤5    | ≤5    | ≤5     | ≤5     |
| Torsional Rigidity           | Nm/arc min | 3~10  | 3                                     | 6      | 14    | 27    | 60    | 140    | 240    |
| Max. Radial Load             | N          | 3~10  | 760                                   | 1,180  | 3,200 | 6,800 | 9,300 | 15,600 | 51,000 |
| Max. Axial Load              | N          | 3~10  | 380                                   | 590    | 1,600 | 3,400 | 4,650 | 7,800  | 25,500 |
| Service Life                 | hr         | 3~10  | 20,000 (4,000 / Continuous Operation) |        |       |       |       |        |        |
| Efficiency                   | %          | 3~10  | ≥ 97                                  |        |       |       |       |        |        |
| Operating Temperature        | °C         | 3~10  | -25°C ~ +90°C                         |        |       |       |       |        |        |
| Lubrication                  |            | 3~10  | VIGO GREASE RE #0                     |        |       |       |       |        |        |
| Degree of Gearbox Protection |            | 3~10  | IP65                                  |        |       |       |       |        |        |
| Mounting Position            |            | 3~10  | Any                                   |        |       |       |       |        |        |
| Noise Level                  | dB         | 3~10  | ≤56                                   | ≤58    | ≤60   | ≤63   | ≤65   | ≤67    | ≤70    |
| Weight ±3%                   | kg         | 3~10  | 0.6                                   | 1.37   | 3.9   | 8     | 14.2  | 29.3   | 39.2   |

| Model code | unit:mm          |                                 |                              |                          |                            |                           |                      |
|------------|------------------|---------------------------------|------------------------------|--------------------------|----------------------------|---------------------------|----------------------|
|            | 44               | 62                              | 90                           | 120                      | 142                        | 180                       | 220                  |
| A1         | 50               | 70                              | 100                          | 130                      | 165                        | 215                       | 250                  |
| A2         | 4.5              | 5.5                             | 6.8                          | 9                        | 11                         | 13                        | 17                   |
| A3         | 13               | 16                              | 22                           | 32                       | 40                         | 55                        | 75                   |
| A4         | 35               | 50                              | 80                           | 110                      | 130                        | 160                       | 180                  |
| A5         | 5                | 6                               | 9                            | 10                       | 10                         | 11.5                      | 12.5                 |
| A6         | M4×P0.7          | M5×P0.8                         | M8×P1.25                     | M10×P1.5                 | M12×P1.75                  | M14×P2.0                  | M16×P2.0             |
| A7         | 58               | 80                              | 116                          | 148                      | 186                        | 238                       | 288                  |
| A8         | 5                | 5                               | 6                            | 10                       | 12                         | 16                        | 20                   |
| A9         | 15               | 18                              | 24.5                         | 35                       | 43                         | 59                        | 79.5                 |
| B1         | 44               | 62                              | 90                           | 120                      | 142                        | 180                       | 220                  |
| B2         | 26               | 36                              | 48                           | 65                       | 92                         | 106                       | 139                  |
| B3         | 5                | 7                               | 10                           | 12                       | 15                         | 20                        | 30                   |
| B4         | 1                | 1                               | 2                            | 3                        | 3                          | 4                         | 5                    |
| B5         | 15               | 20                              | 30                           | 40                       | 65                         | 70                        | 90                   |
| B6         | 20               | 28                              | 36                           | 50                       | 74                         | 82                        | 104                  |
| B7         | 5                | 8                               | 10                           | 12                       | 15                         | 16                        | 20                   |
| B8         | 31.5             | 38                              | 49                           | 61                       | 70                         | 85                        | 93                   |
| B9         | 95               | 115, 123                        | 151.5, 164.5                 | 205                      | 260.5                      | 323.5                     | 367                  |
| B10        | 9                | 11.5                            | 16                           | 19.5                     | 20                         | 23.5                      | 23.5                 |
| C1         | 46, 60, 63, 70   | 70, 75, 90                      | 70, 90, 100<br>115, 145, 165 | 90, 100<br>115, 145, 165 | 145, 165<br>200, 215       | 200, 215<br>235, 265, 300 | 215, 235<br>265, 300 |
| C2         | M4, M5           | M4, M5, M6                      | M6, M8, M10                  | M6, M8, M10              | M8, M12                    | M12, M16                  | M12, M16             |
| C3         | 5, 6.35, 8, (11) | 6.35, 8, 11<br>12, 14, (16, 19) | 14, 16, 19<br>(22, 24)       | 19, 22<br>24, (28, 32)   | 22, 24, 28<br>32, 35, (38) | 38, 42, 48, 55            | 42, 48, 55           |
| C4         | 26               | 33.5, 41.5                      | 46, 59                       | 67                       | 84.5                       | 114.5                     | 117.5                |
| C5         | 30, 40, 50       | 50, 60, 70                      | 50, 70, 80<br>95, 110, 130   | 70, 80<br>95, 110, 130   | 110, 114.3<br>130, 180     | 114.3, 180<br>200, 230    | 180, 200<br>230, 250 |
| C6         | M3               | M5                              | M6                           | M8                       | M10                        | M10                       | M10                  |
| C7         | 46, 55, 60       | 64, 70, 80                      | 92, 110<br>130, 142          | 130, 150                 | 146, 150<br>180, 190       | 182, 200<br>220, 250, 265 | 220, 250, 265        |
| C8         | 37.5             | 41, 49                          | 54.5, 67.5                   | 79                       | 98.5                       | 132.5                     | 135.5                |

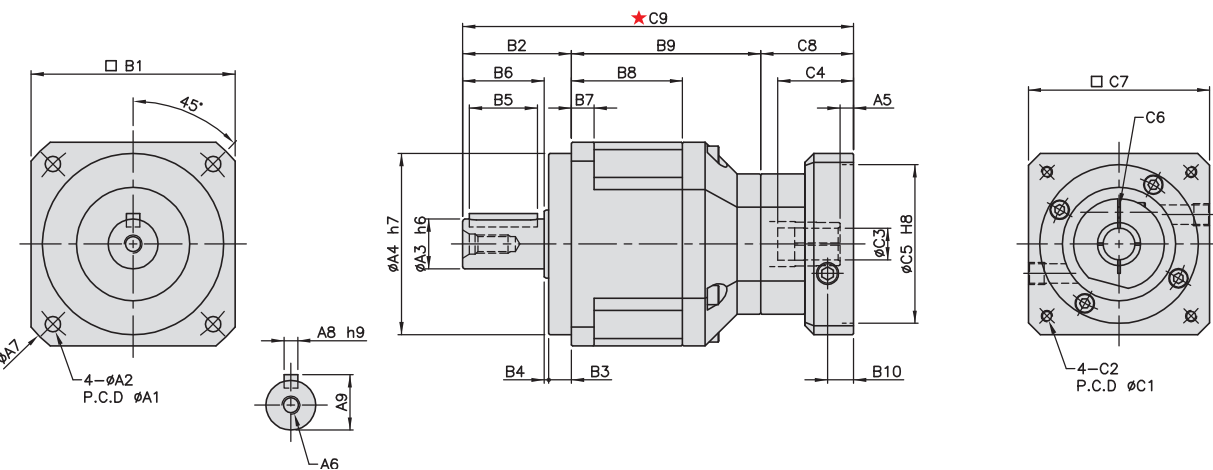
MODEL : KSB

Double Reduction

RATIO : 15, 20, 25, 30, 35, 40, 50, 60, 70, 80, 90, 100



High Precision Planetary Reducer



unit:mm

|   | Model | 62              | 90                             | 120                          | 142                      | 180                       | 220                       |
|---|-------|-----------------|--------------------------------|------------------------------|--------------------------|---------------------------|---------------------------|
|   | code  |                 |                                |                              |                          |                           |                           |
| A | A1    | 70              | 100                            | 130                          | 165                      | 215                       | 250                       |
|   | A2    | 5.5             | 6.8                            | 9                            | 11                       | 13                        | 17                        |
|   | A3    | 16              | 22                             | 32                           | 40                       | 55                        | 75                        |
|   | A4    | 50              | 80                             | 110                          | 130                      | 160                       | 180                       |
|   | A5    | 5               | 6                              | 9                            | 10                       | 10                        | 11.5                      |
|   | A6    | M5×P0.8         | M8×P1.25                       | M10×P1.5                     | M12×P1.75                | M14×P2.0                  | M16×P2.0                  |
|   | A7    | 80              | 116                            | 148                          | 186                      | 238                       | 288                       |
|   | A8    | 5               | 6                              | 10                           | 12                       | 16                        | 20                        |
|   | A9    | 18              | 24.5                           | 35                           | 43                       | 59                        | 79.5                      |
|   | A10   | 18              | 24.5                           | 35                           | 43                       | 59                        | 79.5                      |
| B | B1    | 62              | 90                             | 120                          | 142                      | 180                       | 220                       |
|   | B2    | 36              | 48                             | 65                           | 92                       | 106                       | 139                       |
|   | B3    | 7               | 10                             | 12                           | 15                       | 20                        | 30                        |
|   | B4    | 1               | 2                              | 3                            | 3                        | 4                         | 5                         |
|   | B5    | 20              | 30                             | 40                           | 65                       | 70                        | 90                        |
|   | B6    | 28              | 36                             | 50                           | 74                       | 82                        | 104                       |
|   | B7    | 8               | 10                             | 12                           | 15                       | 16                        | 20                        |
|   | B8    | 38              | 49                             | 61                           | 70                       | 85                        | 93                        |
|   | B9    | 66              | 83.5                           | 108.5                        | 127.5                    | 154                       | 175                       |
|   | B10   | 9               | 11.5                           | 16, 30.5                     | 19.5, 27.5               | 20                        | 23.5                      |
| C | C1    | 46, 60, 63, 70  | 70, 75, 90                     | 70, 90, 100<br>115, 145, 165 | 90, 100<br>115, 145, 165 | 145, 165, 200, 215        | 200, 215<br>235, 265, 300 |
|   | C2    | M4, M5          | M4, M5, M6                     | M6, M8, M10                  | M6, M8, M10              | M8, M12                   | M12, M16                  |
|   | C3    | 5, 6.35, 8,(11) | 6.35, 8, 11<br>12, 14,(16, 19) | 14, 16, 19,(22, 24)          | 19, 22, 24,(28, 32)      | 22, 24, 28<br>32, 35,(38) | 38, 42, 48, 55            |
|   | C4    | 26              | 33.5, 41.5                     | 59                           | 67                       | 84.5                      | 114.5                     |
|   | C5    | 30, 40, 50      | 50, 60, 70                     | 50, 70, 80<br>95, 110, 130   | 70, 80<br>95, 110, 130   | 110, 114.3, 130, 180      | 114.3, 180, 200, 230      |
|   | C6    | M3              | M5                             | M6                           | M8                       | M10                       | M10                       |
|   | C7    | 46, 55, 60      | 64, 70, 80                     | 92, 110, 130, 142            | 130, 150                 | 146, 150, 180, 190        | 182, 200, 220, 250, 265   |
|   | C8    | 37.5            | 41, 49                         | 67.5                         | 79                       | 98.5                      | 132.5                     |
|   | C9    | 139.5           | 172.5, 180.5                   | 241                          | 298.5                    | 358.5                     | 446.5                     |
|   | C10   | 139.5           | 172.5, 180.5                   | 241                          | 298.5                    | 358.5                     | 446.5                     |

■ Mass Moments of Inertia (kg · cm²)

| Ratio | 62   | 90   | 120  | 142  | 180  | 220   |
|-------|------|------|------|------|------|-------|
| 15    | 0.03 | 0.14 | 0.46 | 2.63 | 7.3  | 22.79 |
| 20    | 0.03 | 0.14 | 0.46 | 2.63 | 7.3  | 22.79 |
| 25    | 0.03 | 0.14 | 0.46 | 2.63 | 7.1  | 22.79 |
| 30    | 0.03 | 0.14 | 0.46 | 2.43 | 7.1  | 22.59 |
| 35    | 0.03 | 0.14 | 0.44 | 2.43 | 7.1  | 22.59 |
| 40    | 0.03 | 0.14 | 0.44 | 2.43 | 6.92 | 22.59 |
| 50    | 0.03 | 0.14 | 0.44 | 2.43 | 6.92 | 22.59 |
| 60    | 0.03 | 0.14 | 0.43 | 2.39 | 6.72 | 21.83 |
| 70    | 0.03 | 0.14 | 0.43 | 2.39 | 6.72 | 21.83 |
| 80    | 0.03 | 0.14 | 0.43 | 2.39 | 6.72 | 21.83 |
| 90    | 0.03 | 0.14 | 0.40 | 2.39 | 6.72 | 21.60 |
| 100   | 0.03 | 0.14 | 0.40 | 2.39 | 6.72 | 21.60 |

| Model No.           | Unit | Ratio      | 62                             | 90                                    | 120   | 142   | 180   | 220    |
|---------------------|------|------------|--------------------------------|---------------------------------------|-------|-------|-------|--------|
| Rated Output Torque | Nm   | 15         | 59                             | 165                                   | 335   | 625   | 1206  | 2030   |
|                     |      | 20         | 51                             | 146                                   | 300   | 555   | 1069  | 1804   |
|                     |      | 25         | 48                             | 160                                   | 333   | 618   | 1189  | 2010   |
|                     |      | 30         | 45                             | 151                                   | 311   | 583   | 1118  | 1911   |
|                     |      | 35         | 45                             | 149                                   | 309   | 573   | 1108  | 1870   |
|                     |      | 40         | 43                             | 143                                   | 298   | 553   | 1070  | 1824   |
|                     |      | 50         | 48                             | 160                                   | 333   | 618   | 1189  | 2010   |
|                     |      | 60         | 45                             | 151                                   | 311   | 583   | 1118  | 1911   |
|                     |      | 70         | 45                             | 149                                   | 309   | 573   | 1108  | 1870   |
|                     |      | 80         | 43                             | 143                                   | 298   | 553   | 1070  | 1824   |
| Max. Output Torque  | Nm   | 90         | 44                             | 145                                   | 278   | 516   | 993   | 1694   |
|                     |      | 100        | 43                             | 141                                   | 294   | 549   | 1059  | 1779   |
|                     |      | 15~100     | 3 Times of Rated Output Torque |                                       |       |       |       |        |
|                     |      | rpm        | 15~100                         | 5,000                                 | 4,000 | 4,000 | 3,000 | 2,000  |
|                     |      | rpm        | 15~100                         | 10,000                                | 8,000 | 8,000 | 6,000 | 4,000  |
|                     |      | arc min    | 15~100                         | ≤3                                    |       |       |       |        |
|                     |      | arc min    | 15~100                         | ≤5                                    | ≤5    | ≤5    | ≤5    | ≤5     |
|                     |      | arc min    | 15~100                         | ≤7                                    | ≤7    | ≤7    | ≤7    | ≤7     |
|                     |      | Nm/arc min | 15~100                         | 6                                     | 14    | 27    | 60    | 240    |
|                     |      | N          | 15~100                         | 1,180                                 | 3,200 | 6,800 | 9,300 | 51,000 |
| Max. Axial Load     | N    | 15~100     | 590                            | 1,600                                 | 3,400 | 4,650 | 7,800 | 25,500 |
|                     |      | hr         | 15~100                         | 20,000 (4,000 / Continuous Operation) |       |       |       |        |
|                     |      | %          | 15~100                         | ≥94%                                  |       |       |       |        |
|                     |      | °C         | 15~100                         | -25°C ~ +90°C                         |       |       |       |        |
|                     |      |            | 15~100                         | VIGO GREASE RE #0                     |       |       |       |        |
|                     |      |            | 15~100                         | IP65                                  |       |       |       |        |
|                     |      |            | 15~100                         | Any                                   |       |       |       |        |
|                     |      | dB         | 15~100                         | ≤58                                   | ≤60   | ≤63   | ≤65   | ≤67    |
|                     |      | kg         | 15~100                         | 1.73                                  | 4.6   | 9.42  | 20.5  | 39.14  |
|                     |      |            | 15~100                         | ≤70                                   |       |       |       |        |